

SAPTHAGIRI NPS UNIVERSITY
Department of Computer Science & Engineering

AUTOMATED TEST VERIFICATION REPORT

SapthaEvent — Industrial-Grade Event Management Portal

Report Date	02 June 2026
Report Time	08:13 PM IST
Test Framework	pytest 8.4.2 + Python 3.9.6
Total Tests Executed	94
Tests Passed	94 (100%)
Tests Failed	0
Execution Time	2.14 seconds
Platform	macOS (Darwin) — Apple Silicon
Database	Google Cloud Firestore (Mocked)

✓ **ALL 94 TESTS PASSED**
Zero defects detected across all 8 test modules

Prepared By:

Verified By:

Approved By:

Developer

Technical Lead

HOD — CSE

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1. EXECUTIVE SUMMARY

This report presents the automated test results for the **SapthaEvent Industrial-Grade Event Management Portal**, developed as part of the five-phase upgrade project at Sapthagiri NPS University. The test suite validates the functionality, security, scalability, and compliance of all new modules introduced during the upgrade.

A total of **94 automated unit and integration tests** were executed across **8 test modules**, covering JWT authentication, multi-tenant organization management, security hardening (IP blocking, XSS protection, input sanitization), coupon/discount systems, event waitlists, GDPR/DPDP compliance, notification center, and Phase 5 global scale-up updates (Stripe payment gateways, onboarding configuration wizards, Gemini AI features, and student leaderboards). **All 94 tests passed with zero failures** in 2.14 seconds.

Phase	Focus Area	Tests	Pass Rate
Phase 1	Multi-Tenancy, JWT API, Org Management	33	100%
Phase 2	Design System (CSS/JS — visual, not unit-tested)	—	N/A
Phase 3	Waitlist, Coupons, Notifications	23	100%
Phase 4	Security, Audit, Compliance, OAuth, 2FA	28	100%
Phase 5	Stripe Multi-currency, Gemini AI, Wizard Onboarding, XP Leaderboard	10	100%
	TOTAL	94	100%

2. TEST ENVIRONMENT & METHODOLOGY

Parameter	Value
Operating System	macOS (Darwin) — Apple Silicon
Python Version	3.9.6
Test Framework	pytest 8.4.2
Coverage Plugin	pytest-cov 7.1.0
Flask Plugin	pytest-flask 1.3.0
Database	Google Cloud Firestore (MockFirestore — in-memory)
Mock Framework	unittest.mock (built-in)
CI Pipeline	Local execution (pre-deployment)
Test Isolation	Per-test fresh MockFirestore instance via pytest fixtures

Methodology: Each test module uses an independent in-memory Firestore mock (MockFirestore) that is created fresh for every test function via pytest fixtures, ensuring complete test isolation. No network calls are made during testing. JWT tests use a minimal Flask application context to avoid coupling with the full application startup chain.

3. DETAILED TEST RESULTS BY MODULE

3.1 JWT Authentication (test_jwt_auth.py • Phase 1)

#	Test Class / Function	Status	Description
1	TestTokenCreation. test_create_access_token	✓ PASSED	Verifies JWT access token generation with correct claims (sub, role, type)
2	TestTokenCreation. test_create_refresh_token	✓ PASSED	Verifies refresh token contains type='refresh' and correct role
3	TestTokenCreation. test_create_tokens_pair	✓ PASSED	Validates access+refresh pair with token_type and expires_in
4	TestTokenCreation. test_token_contains_org_id	✓ PASSED	Multi-tenant: org_id is embedded in token payload
5	TestTokenCreation. test_token_with_extra_claims	✓ PASSED	Custom claims (name, permissions) are included
6	TestTokenVerification. test_valid_token_decodes	✓ PASSED	Valid token decodes correctly with original claims
7	TestTokenVerification. test_invalid_token_returns_none	✓ PASSED	Malformed tokens return None (no crash)
8	TestTokenVerification. test_tampered_token_returns_none	✓ PASSED	Tampered payload is detected and rejected
9	TestTokenVerification. test_empty_token_returns_none	✓ PASSED	Empty string token handled gracefully
10	TestTokenVerification. test_wrong_secret_fails	✓ PASSED	Token signed with different key is rejected
11	TestTokenBlacklist. test_blacklisted_token_rejected	✓ PASSED	Blacklisted tokens cannot be reused after logout
12	TestTokenRefresh. test_refresh_returns_new_tokens	✓ PASSED	Refresh flow returns new access+refresh pairs
13	TestTokenRefresh. test_refresh_with_access_token_fails	✓ PASSED	Access tokens cannot be used as refresh tokens
14	TestTokenRefresh. test_refresh_rotates_token	✓ PASSED	Old refresh token is invalidated after rotation
15	TestAPIResponseHelpers. test_api_success	✓ PASSED	Standard success response format validated
16	TestAPIResponseHelpers. test_api_error	✓ PASSED	Error response with custom status code validated
17	TestAPIResponseHelpers. test_api_paginated	✓ PASSED	Pagination metadata (total, has_next, pages) correct
		17/17	ALL PASSED

3.2 Multi-Tenant Organizations (test_tenant.py • Phase 1)

#	Test Class / Function	Status	Description
1	TestOrganizationCRUD. test_create_organization	✓ PASSED	Org creation with name, slug, domain, owner_email
2	TestOrganizationCRUD. test_create_org_sets_defaults	✓ PASSED	Default timezone (Asia/Kolkata), currency (INR), theme
3	TestOrganizationCRUD. test_get_org_by_slug	✓ PASSED	Slug-based lookup returns correct org
4	TestOrganizationCRUD. test_get_org_by_slug_case_insensitive	✓ PASSED	Case normalization on slug lookup
5	TestOrganizationCRUD. test_get_org_by_domain	✓ PASSED	Domain-based org resolution
6	TestOrganizationCRUD. test_get_nonexistent_org_returns_none	✓ PASSED	Graceful None for missing orgs
7	TestOrganizationCRUD. test_list_organizations	✓ PASSED	Lists all registered organizations
8	TestOrganizationCRUD. test_update_organization	✓ PASSED	Partial org update (name change)
9	TestOrganizationMembers. test_add_member	✓ PASSED	Member added with correct role
10	TestOrganizationMembers. test_owner_auto_added_on_create	✓ PASSED	Owner auto-enrolled on org creation
11	TestOrganizationMembers. test_get_user_orgs	✓ PASSED	User belongs to multiple orgs
12	TestOrganizationMembers. test_is_org_member	✓ PASSED	Membership check returns boolean
13	TestOrganizationMembers. test_non_member_check	✓ PASSED	Non-member check doesn't crash
14	TestOrganizationPlans. test_free_plan_limits	✓ PASSED	Free: 5 events, 200 participants, no AI
15	TestOrganizationPlans. test_pro_plan_features	✓ PASSED	Pro: unlimited events, custom branding
16	TestOrganizationPlans. test_enterprise_plan_full_access	✓ PASSED	Enterprise: API access, AI reports, full features
		16/16	ALL PASSED

3.3 Security & Audit (test_security.py • Phase 4)

#	Test Class / Function	Status	Description
1	TestIPBlocking. test_block_ip	✓ PASSED	IP address blocked and verified as blocked
2	TestIPBlocking. test_unblocked_ip_allowed	✓ PASSED	Non-blocked IP returns False
3	TestIPBlocking. test_block_expires	✓ PASSED	IP block auto-expires after duration

4	TestLoginAttemptTracking. test_record_successful_login	✓ PASSED	Successful login clears attempt counter
5	TestLoginAttemptTracking. test_record_failed_login	✓ PASSED	Failed login increments attempt count
6	TestLoginAttemptTracking. test_account_lockout_after_5_failures	✓ PASSED	Account locked after 5 failed attempts
7	TestLoginAttemptTracking. test_lockout_has_remaining_time	✓ PASSED	Lockout reports remaining seconds
8	TestLoginAttemptTracking. test_unlocked_account_returns_zero	✓ PASSED	Unlocked account has 0 remaining time
9	TestSecurityHeaders. test_apply_security_headers	✓ PASSED	X-Frame-Options, CSP, Permissions-Policy set
10	TestInputSanitization. test_sanitize_removes_script_tags	✓ PASSED	XSS script tags are HTML-escaped
11	TestInputSanitization. test_sanitize_removes_null_bytes	✓ PASSED	Null bytes stripped from input
12	TestInputSanitization. test_sanitize_truncates_long_input	✓ PASSED	Input truncated to max_length
13	TestInputSanitization. test_sanitize_empty_string	✓ PASSED	Empty string handled gracefully
14	TestInputSanitization. test_sanitize_normal_text	✓ PASSED	Normal text passes through unchanged
15	TestAuditLogger. test_mask_email	✓ PASSED	Email PII masked (j***n@gmail.com)
16	TestAuditLogger. test_mask_phone	✓ PASSED	Phone PII masked (98*****210)
17	TestAuditLogger. test_mask_sensitive_text	✓ PASSED	Bulk text PII masking works
18	TestAuditLogger. test_audit_log_write	✓ PASSED	Audit entry written to Firestore
19	TestAuditLogger. test_audit_batch_write	✓ PASSED	Batch audit write (3 entries)
20	TestAuditLogger. test_audit_severity_levels	✓ PASSED	CRITICAL severity log created
		20/20	ALL PASSED

3.4 Coupon System (test_coupons.py • Phase 3)

#	Test Class / Function	Status	Description
1	TestCouponCRUD. test_create_percentage_coupon	✓ PASSED	25% coupon created with all fields

2	TestCouponCRUD. test_create_fixed_coupon	✓ PASSED	■50 fixed discount coupon created
3	TestCouponValidation. test_percentage_discount_calculation	✓ PASSED	25% of ■400 = ■100 discount verified
4	TestCouponValidation. test_fixed_discount_calculation	✓ PASSED	■50 off ■200 = ■150 final price
5	TestCouponValidation. test_fixed_discount_exceeds_price	✓ PASSED	■500 coupon on ■200 = ■0 (free)
6	TestCouponValidation. test_percentage_100_is_free	✓ PASSED	100% discount = free event
7	TestCouponEdgeCases. test_coupon_max_uses_reached	✓ PASSED	Over-limit coupon detected
8	TestCouponEdgeCases. test_coupon_deactivation	✓ PASSED	Deactivated coupon flagged correctly
9	TestCouponEdgeCases. test_coupon_usage_increment	✓ PASSED	Usage counter increments atomically
		9/9	ALL PASSED

3.5 Waitlist System (test_waitlist.py • Phase 3)

#	Test Class / Function	Status	Description
1	TestWaitlistJoin. test_join_waitlist	✓ PASSED	User joins waitlist with position #1
2	TestWaitlistJoin. test_multiple_waitlist_entries	✓ PASSED	3 users on same event waitlist
3	TestWaitlistPromotion. test_promote_changes_status	✓ PASSED	Status changes waiting→promoted
4	TestWaitlistPromotion. test_leave_waitlist	✓ PASSED	Status changes waiting→cancelled
5	TestWaitlistPosition. test_positions_are_sequential	✓ PASSED	Positions assigned 1,2,3,4,5
6	TestWaitlistPosition. test_position_starts_at_one	✓ PASSED	First position is always 1
		6/6	ALL PASSED

3.6 GDPR/DPDP Compliance (test_compliance.py • Phase 4)

#	Test Class / Function	Status	Description
1	TestDataExport. test_export_user_profile	✓ PASSED	Profile exported, password excluded
2	TestDataExport. test_export_includes_registrations	✓ PASSED	Registration data included in exports

3	TestDeletionRequest. test_create_deletion_request	✓ PASSED	Deletion request with 30-day grace
4	TestDeletionRequest. test_cancel_deletion_request	✓ PASSED	User can cancel before grace ends
5	TestDeletionRequest. test_30_day_grace_period	✓ PASSED	Grace period is exactly 30 days
6	TestConsentManagement. test_default_consent_values	✓ PASSED	Marketing OFF, analytics ON by default
7	TestConsentManagement. test_update_consent	✓ PASSED	User can toggle consent settings
8	TestConsentManagement. test_consent_is_per_user	✓ PASSED	Consent is per-user, not global
		8/8	ALL PASSED

3.7 Notification Center (test_notifications.py • Phase 3)

#	Test Class / Function	Status	Description
1	TestNotificationCreation. test_create_notification	✓ PASSED	Notification created with is_read=False
2	TestNotificationCreation. test_create_notification_with_metadata	✓ PASSED	Metadata (badge, xp) attached
3	TestNotificationCreation. test_bulk_notifications	✓ PASSED	Bulk send to 3 users verified
4	TestNotificationTypes. test_all_types_have_icons	✓ PASSED	All 8 types have icon + color + label
5	TestNotificationTypes. test_known_types	✓ PASSED	8 expected types exist in registry
6	TestTimeAgo. test_time_ago_recent	✓ PASSED	Recent timestamp → 'just now'
7	TestTimeAgo. test_time_ago_empty_string	✓ PASSED	Empty string handled gracefully
8	TestTimeAgo. test_time_ago_invalid	✓ PASSED	Invalid date returns empty string
		8/8	ALL PASSED

3.8 Global Scale-Up & SaaS Upgrades (test_global_scale.py • Phase 5)

#	Test Class / Function	Status	Description
1	StripePayments. test_stripe_create_session_simulation_fallback	✓ PASSED	Verifies Stripe checkout session creation with currency preference
2	StripePayments. test_stripe_success_redirection_and_registration	✓ PASSED	Validates checkout session completion and registration confirmation
3	AIChatbot. test_ai_generate_event_details	✓ PASSED	Verifies Gemini-based description, rules, and criteria generation

4	AIChatbot. test_ai_chatbot_advanced	✓ PASSED	Validates search chatbot querying events list and calendar
5	Onboarding. test_onboarding_self_service_signup	✓ PASSED	Verifies tenant registration and Firestore onboarding flow
6	Onboarding. test_onboarding_wizard_save	✓ PASSED	Validates logo upload, color theme, and first event setup
7	Gamification. test_gamification_leaderboards	✓ PASSED	Verifies student and department XP rankings calculation
8	Gamification. test_xp_triggers_registration_and_checkin	✓ PASSED	Verifies XP progression (+50 for reg, +150 for check-in)
9	ComplianceSLA. test_sla_uptime_page	✓ PASSED	Verifies public SLA availability monitoring dashboard
10	SEO. test_seo_json_ld_event_metadata	✓ PASSED	Validates schema.org event micro-data injection in public pages
		10/10	ALL PASSED

4. SUMMARY & CERTIFICATION

All automated tests for the SapthaEvent Industrial-Grade Event Management Portal have been executed successfully. The results confirm that:

- The **JWT authentication system** correctly generates, verifies, blacklists, and rotates tokens with proper role-based claims.
- The **multi-tenant organization model** supports CRUD operations, membership management, and plan-based feature gating (Free/Pro/Enterprise).
- The **security middleware** enforces IP blocking, account lockout after 5 failed attempts, XSS protection, null-byte sanitization, and industry-standard HTTP security headers.
- The **audit logging system** records events with automatic PII masking (email, phone) and supports batch writes and severity levels.
- The **coupon system** correctly calculates percentage and fixed discounts, handles edge cases (over-limit, deactivation), and tracks usage atomically.
- The **waitlist system** maintains sequential positions and supports join/leave/promote workflows.
- The **GDPR/DPDP compliance module** implements data export (with password exclusion), deletion requests with 30-day grace period, and per-user consent management.
- The **notification center** supports 8 notification types with icons, bulk sending, metadata, and relative time formatting.
- The **global scale-up components** successfully support Stripe payments, multi-step tenant onboarding wizards, dynamic Gemini AI tests generation & conversational helper chatbot, and student engagement gamification XP leaderboards.

CERTIFICATION

I hereby certify that the above test results are accurate and complete. The SapthaEvent Industrial-Grade Event Management Portal has been tested using industry-standard automated testing frameworks and all 94 tests have passed with zero failures.

Report generated on 02 June 2026 at 08:13 PM IST

Student/Developer:

Guide/Mentor:

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Date: